

SIMATS ENGINEERING
C01 - ASSESSMENT TOOL 1

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COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1103

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Girija B/192311044 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
GIRIJA B

1. Construct a Concept Map

Given:

Requirement Analysis → _____ → Coding → Testing → _____ → Maintenance

Answer:

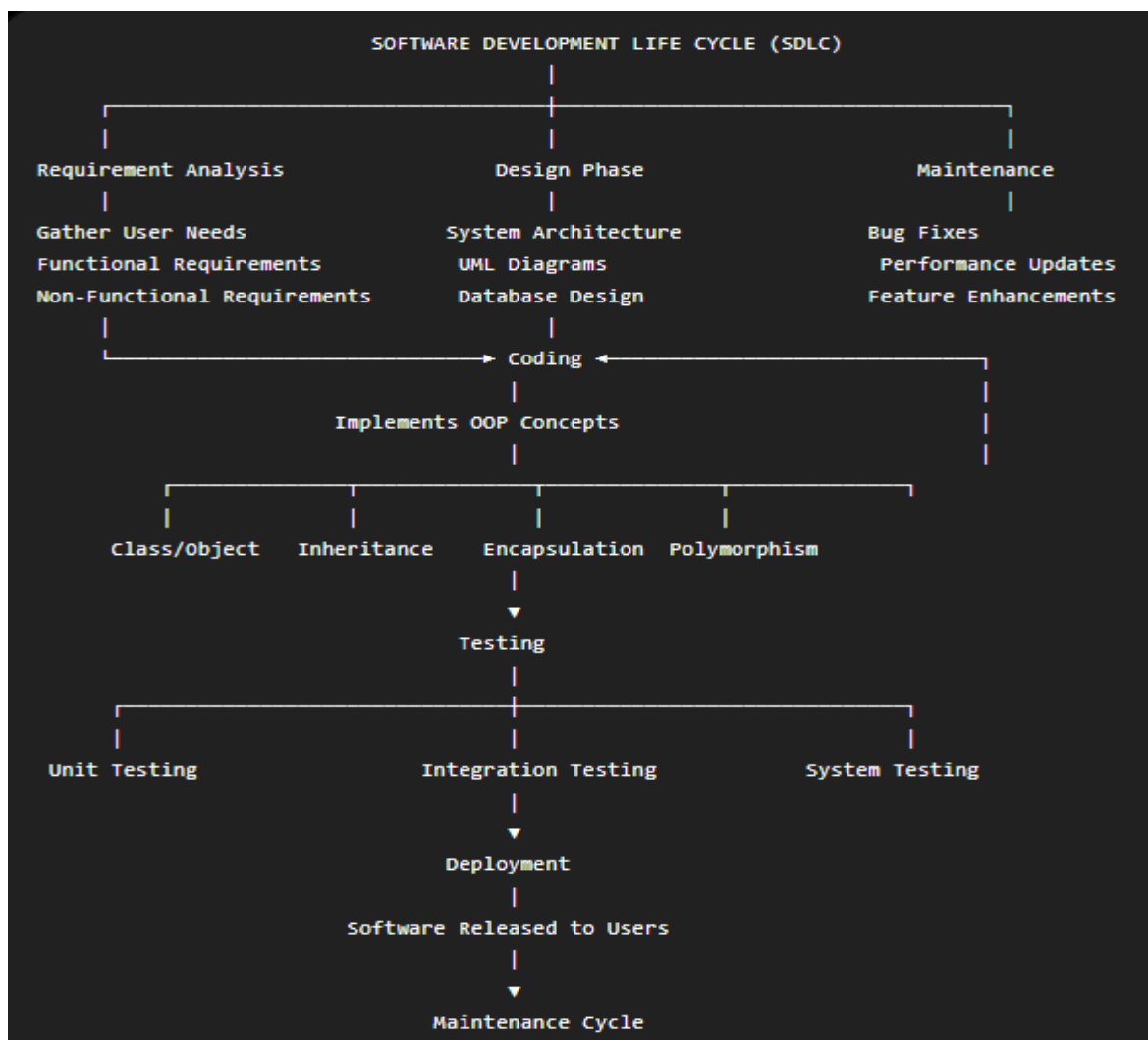
Requirement Analysis → Design → Coding → Testing → Deployment → Maintenance

Explanation

- Design: Converts the software requirements into system architecture, database design, user interface, and module design before coding begins.
- Deployment: Releases the tested software to the end users or production environment for actual use.

Analysis

- Provides a structured Software Development Life Cycle (SDLC).
- Reduces project risks by following defined phases.
- Improves software quality, reliability, and maintainability.
- Ensures each phase is completed before moving to the next stage.
- Helps in delivering software that meets user requirements efficiently.



2. Construct a Concept Map

Given:

Object-Oriented Principles → Abstraction → → Inheritance →

Answer:

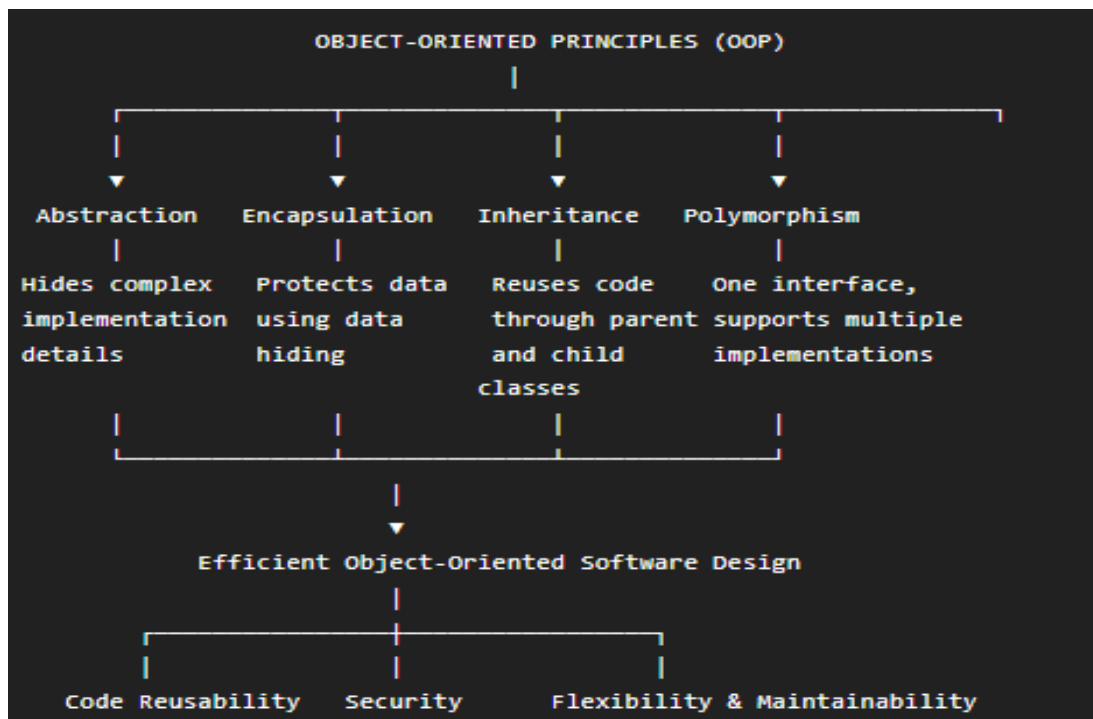
- Encapsulation
- Polymorphism

Explanation:

- Encapsulation protects data from unauthorized access.
- Polymorphism allows one interface to support multiple behaviors.

Analysis:

- Improves security and flexibility.
- Enhances code reusability.
- Supports efficient software design.



3. Construct a Concept Map

Given:

Class → Blueprint

Object → Instance

Method → _____

Attribute → _____

Answer:

The missing concepts are:

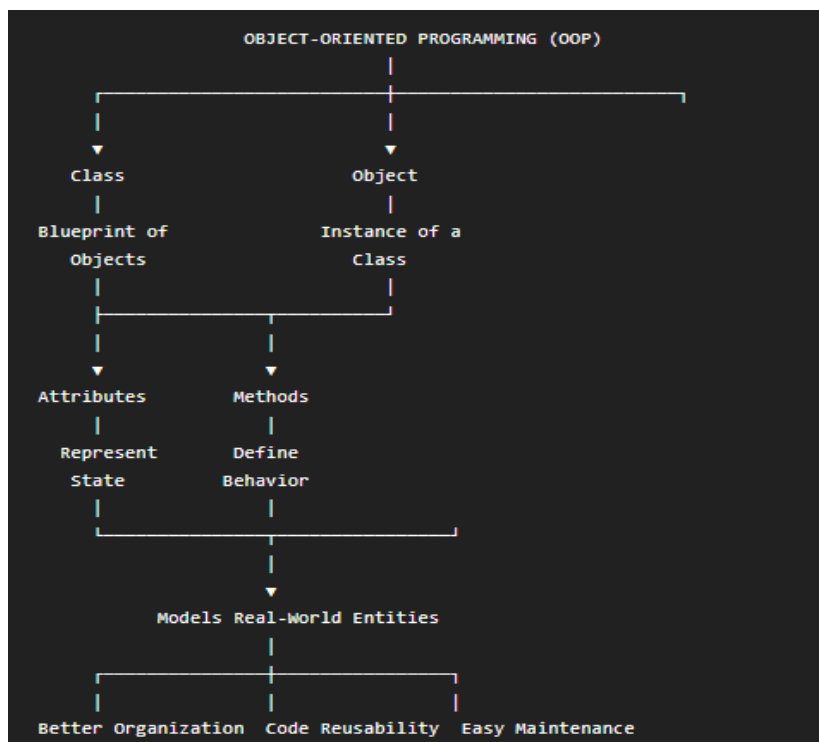
- Behavior
- State

Explanation:

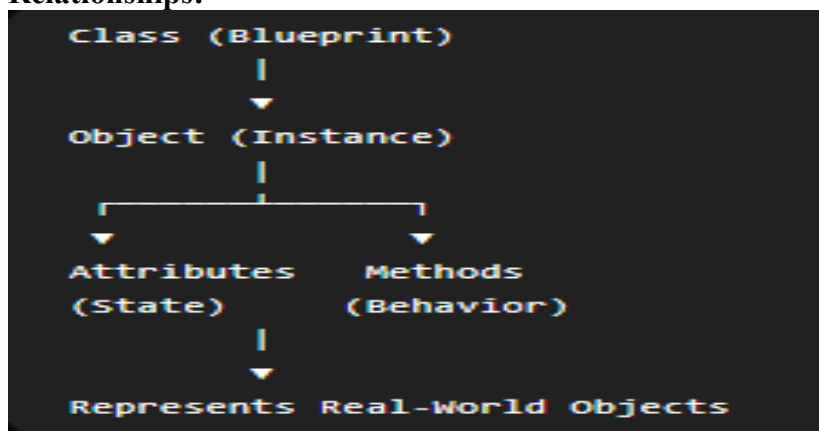
- Methods define the behavior of objects.
- Attributes represent the state of objects.

Analysis:

- Helps model real-world entities.
- Defines object characteristics and actions.
- Improves system organization.



Relationships:



4. A Concept Map Shows

Given:

Software Quality → **Reliability** → _____ → **Maintainability**

Answer:

The missing concept is:

- Usability

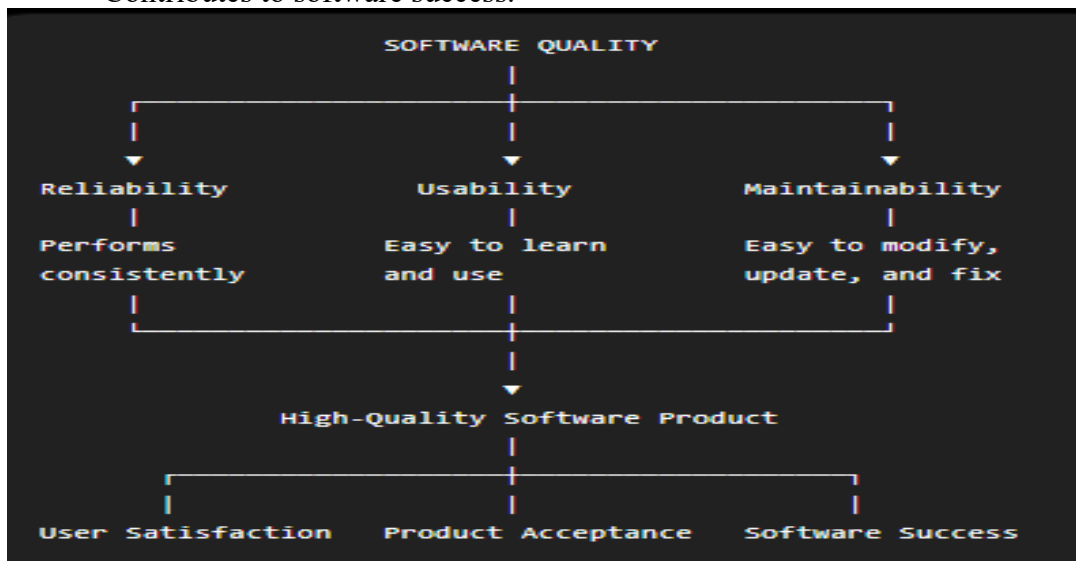
Explanation:

- Usability measures how easily users interact with software.
- Reliable software performs consistently.
- Maintainable software is easy to update.

Analysis:

- Improves user satisfaction.

- Enhances product acceptance.
- Contributes to software success.



5. UML Concept Map

Given:

UML → Structural Diagrams → _____ → Behavioral Diagrams → Activity Diagram → Interaction Diagrams → _____

Answer:

The missing concepts are:

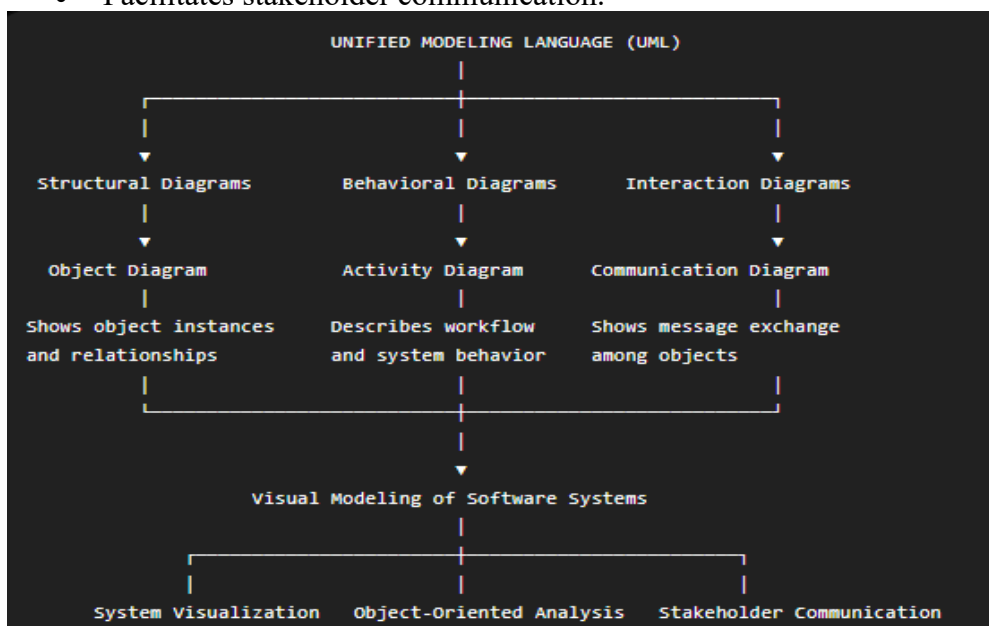
- Object Diagram
- Communication Diagram

Explanation:

- Object diagrams represent object instances and relationships.
- Communication diagrams show message exchange among objects.

Analysis:

- Improves system visualization.
- Supports object-oriented analysis.
- Facilitates stakeholder communication.



6. Given:

Requirements ✓ → Design ✓ → Coding ✗ → Testing ✗ → Deployment ✗

Explanation:

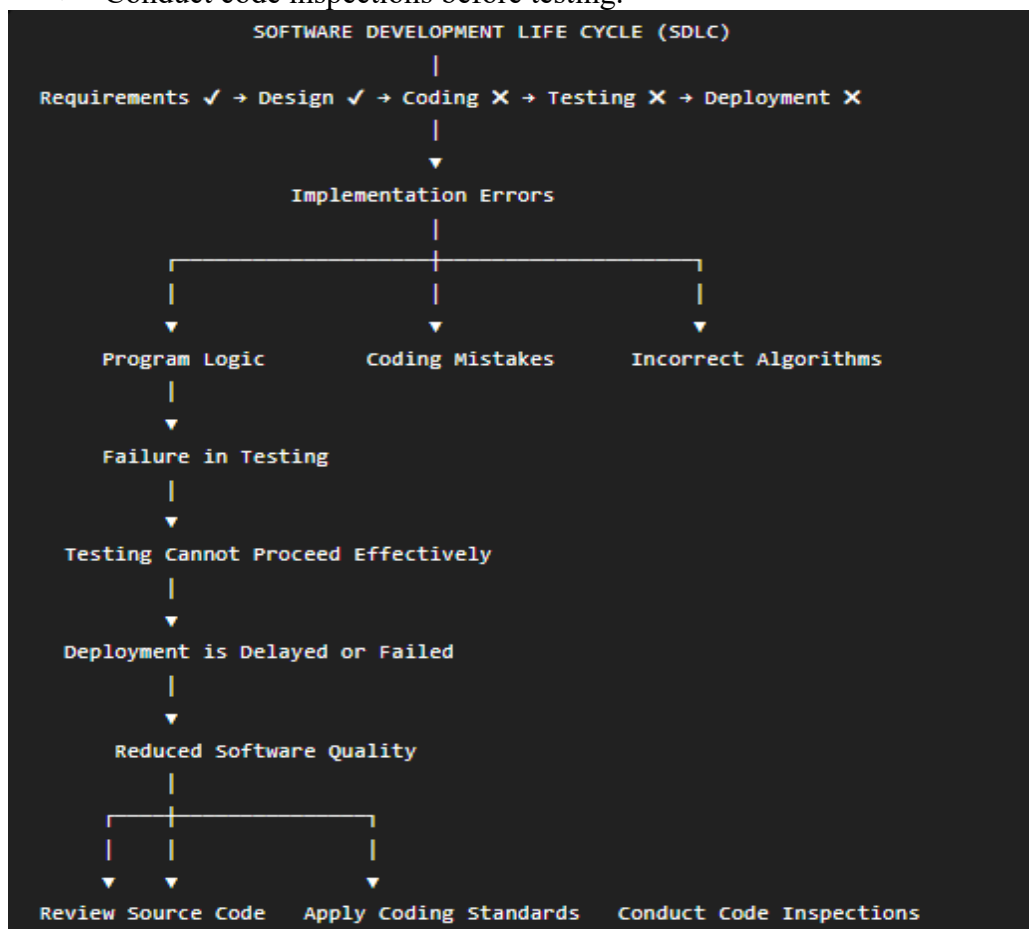
- Failure occurs during the Coding phase.
- Program logic contains implementation errors.

Analysis:

- Errors affect subsequent phases.
- Testing cannot proceed effectively.
- Software quality decreases.

Solution:

- Review source code.
- Apply coding standards.
- Conduct code inspections before testing.



Analysis

- Errors in the **Coding** phase affect all subsequent SDLC phases.
- Testing becomes ineffective due to faulty implementation.
- Software quality, reliability, and user satisfaction decrease.
- Early detection through code reviews reduces project risks.

7. Construct a Concept Map

Given:

Problem Identification → _____ → Design → Implementation → _____

Answer:

The missing concepts are:

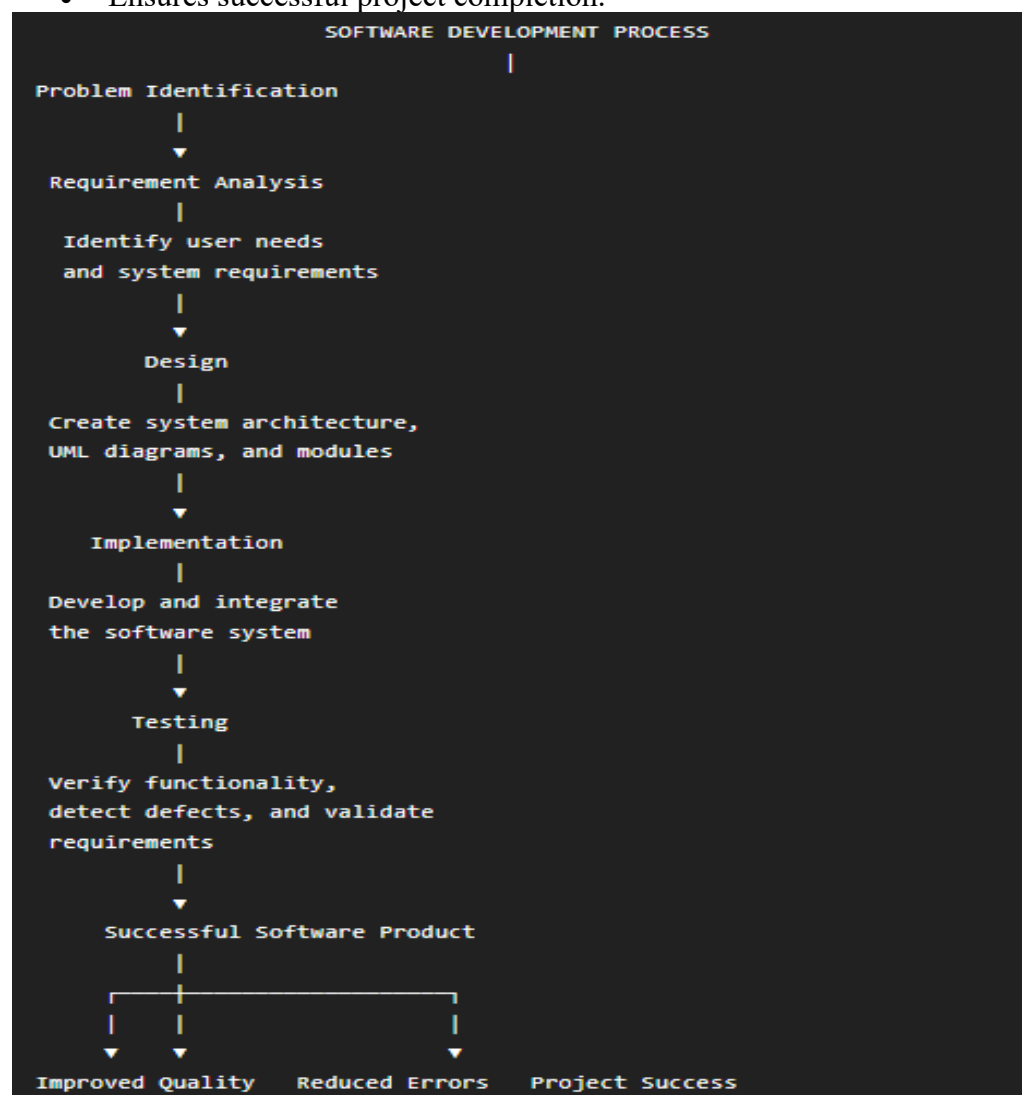
- Requirement Analysis
- Testing

Explanation:

- Requirement Analysis identifies user needs and system requirements.
- Testing verifies whether the implemented system meets the specified requirements.

Analysis:

- Improves software quality.
- Reduces development errors.
- Ensures successful project completion.



Analysis

- Improves **software quality** through systematic development.
- Reduces **development errors** by validating requirements and testing the implementation.
- Ensures **successful project completion** by following a structured development process.

8. Construct a Concept Map

Given:

Object-Oriented Features → Class → _____ → Message Passing → _____

Answer:

The missing concepts are:

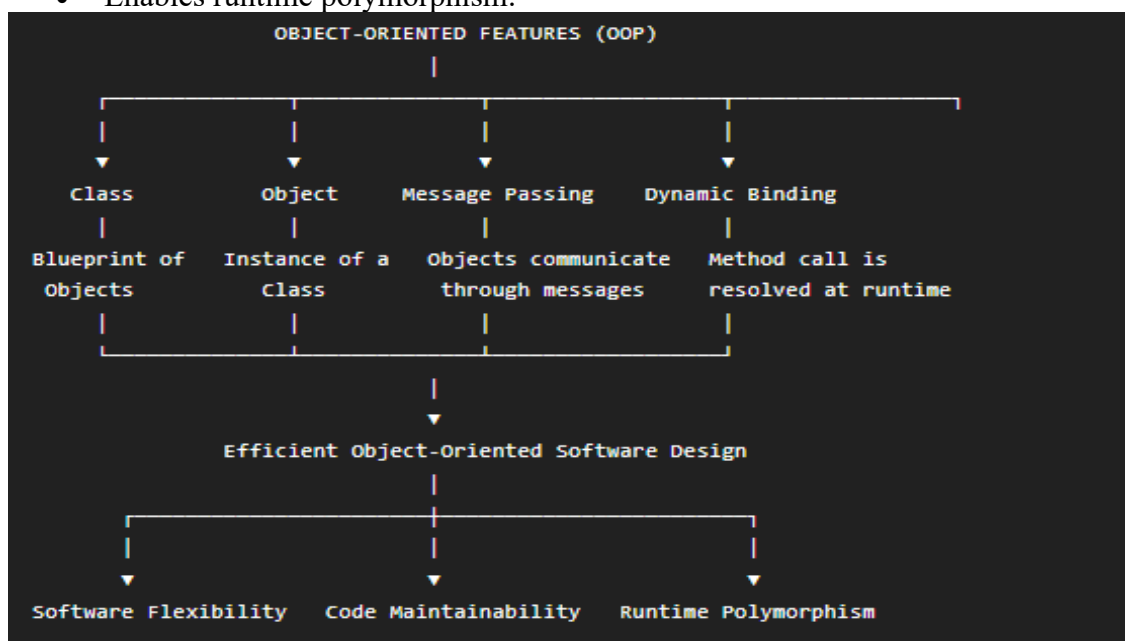
- Object
- Dynamic Binding

Explanation:

- Objects are instances of classes.
- Dynamic binding allows method calls to be resolved at runtime.

Analysis:

- Supports flexibility in software design.
- Improves code maintainability.
- Enables runtime polymorphism.



Analysis

- Supports **flexibility** in software design through dynamic method resolution.
- Improves **code maintainability** by promoting modular and reusable components.
- Enables **runtime polymorphism**, allowing objects to exhibit different behaviors through a common interface.
- Enhances communication between objects using **message passing**.

9. Construct a Concept Map

Given:

Software Development Models → Waterfall → _____ → Spiral → _____

Answer:

The missing concepts are:

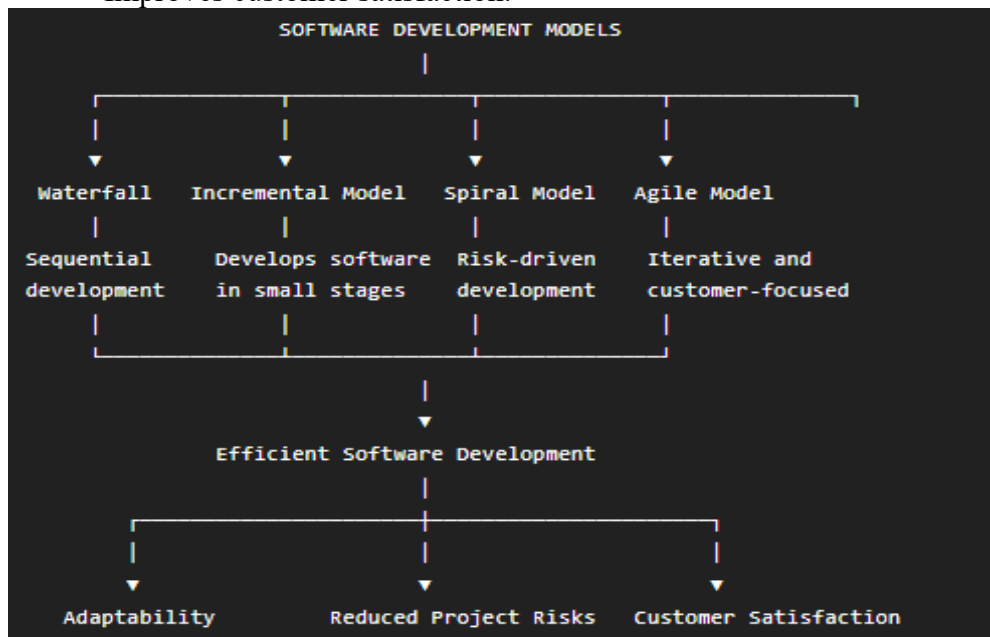
- Incremental Model
- Agile Model

Explanation:

- Incremental development delivers software in stages.
- Agile supports iterative development with continuous customer feedback.

Analysis:

- Enhances adaptability.
- Reduces project risks.
- Improves customer satisfaction.



Analysis

- Enhances **adaptability** by accommodating changing requirements.
- Reduces **project risks** through iterative planning and risk assessment.
- Improves **customer satisfaction** with continuous feedback and frequent software delivery.
- Helps organizations choose the most suitable development model based on project needs.

10. Construct a Concept Map

Given:

Class Diagram → Class → _____ → Association → _____

Answer:

The missing concepts are:

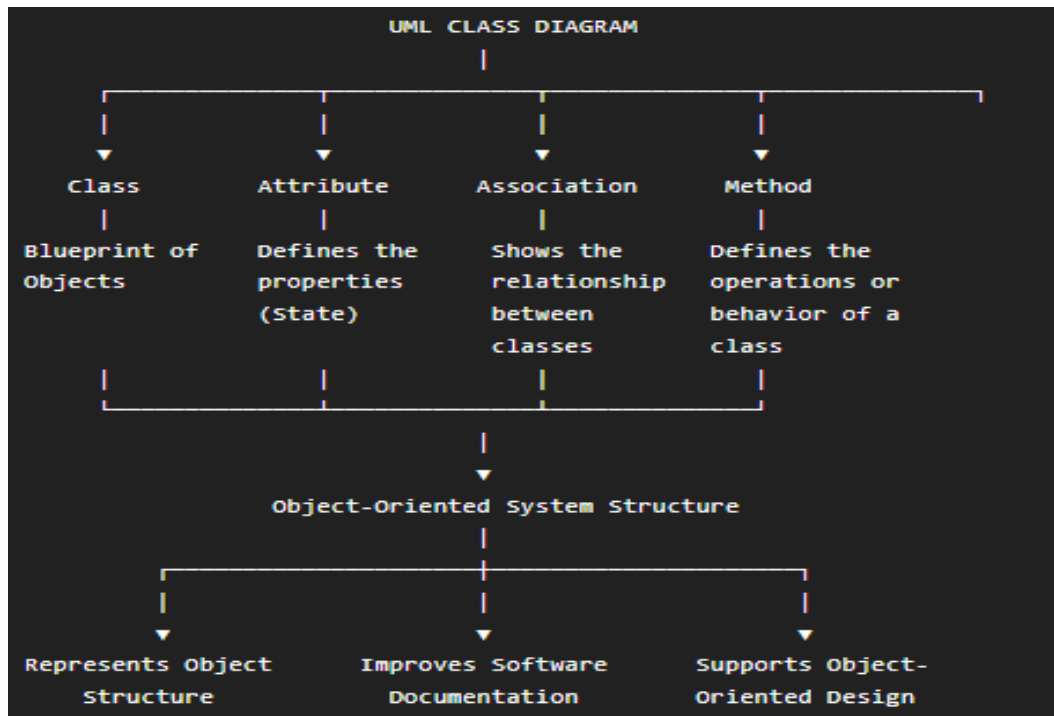
- Attribute
- Method

Explanation:

- Attributes define the properties of a class.
- Methods define the operations performed by a class.

Analysis:

- Represents the **object structure** of a software system effectively.
- Improves **software documentation** by clearly defining classes, attributes, methods, and relationships.
- Supports **object-oriented design** by providing a visual representation of the system architecture.
- Enhances code organization, maintainability, and communication among developers.



11. Construct a Concept Map

Given:

Software Engineering Principles → Modularity → _____ → Reusability → _____

Answer:

The missing concepts are:

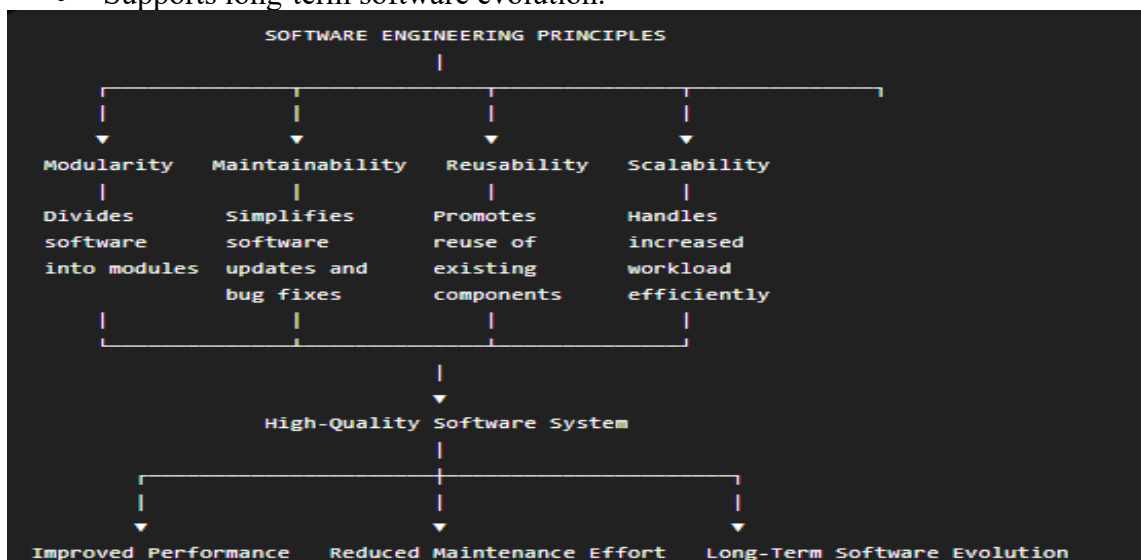
- Maintainability
- Scalability

Explanation:

- Maintainability simplifies software updates.
- Scalability enables systems to handle increased workload efficiently.

Analysis:

- Improves software performance.
- Reduces maintenance effort.
- Supports long-term software evolution.



12. Given:

Requirement Analysis ✓ → System Design ✓ → Coding ✓ → Integration ✗ → Deployment ✗

Explanation:

- Failure occurs during the Integration phase.
- Individual modules fail to communicate correctly.

Analysis:

- Causes functional inconsistencies.
- Delays software deployment.
- Increases debugging effort.

Solution:

- Perform integration testing.
- Verify module interfaces.
- Resolve compatibility issues before deployment.

